

Student: Jonatan Rabjerg
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Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 9

A probability density function of a random variable is given by $f(x) = \frac{x}{32} - \frac{1}{32}$ on the interval $[1, 9]$. Find the expected value, the variance, and the standard deviation.

First, find the expected value. If X is a continuous random variable with probability density function f on $[a, b]$, then the expected value of X is

$\mu = \int_a^b x f(x) dx$. Multiply the probability function by x and integrate over the domain of the function.

$$\mu = \int_1^9 \left(\frac{x^2}{32} - \frac{x}{32} \right) dx$$

Rewrite the integral, factoring out constants, and then find the antiderivatives of x^2 and x .

$$\begin{aligned} \int_1^9 \left(\frac{x^2}{32} - \frac{x}{32} \right) dx &= \frac{1}{32} \int_1^9 x^2 dx - \frac{1}{32} \int_1^9 x dx \\ &= \frac{1}{32} \left(\frac{x^3}{3} \right) \Big|_1^9 - \frac{1}{32} \left(\frac{x^2}{2} \right) \Big|_1^9 \end{aligned}$$

Now factor out constants, substitute the limits of integration, and simplify.

$$\begin{aligned} \frac{1}{32} \left(\frac{x^3}{3} \right) \Big|_1^9 - \frac{1}{32} \left(\frac{x^2}{2} \right) \Big|_1^9 &= \frac{1}{96} (9^3 - 1^3) - \frac{1}{64} (9^2 - 1^2) \\ &\approx 6.33 \end{aligned}$$

The expected value is $\mu = 6.33$. Next, find the variance. If X is a continuous random variable with probability density function f on $[a, b]$, then

the variance of X is $\text{Var}(X) = \int_a^b (x - \mu)^2 f(x) dx$. It can also be found using the alternative formula $\text{Var}(X) = \int_a^b x^2 f(x) dx - \mu^2$.

We will use the alternative formula.

$$\text{Var}(X) = \int_1^9 \left(\frac{x^3}{32} - \frac{x^2}{32} \right) dx - 6.33^2$$

Rewrite the integral, factoring out constants, and then find the antiderivatives of x^3 and x^2 .

$$\begin{aligned} \int_1^9 \left(\frac{x^3}{32} - \frac{x^2}{32} \right) dx - 6.33^2 &= \frac{1}{32} \int_1^9 x^3 dx - \frac{1}{32} \int_1^9 x^2 dx - 6.33^2 \\ &= \frac{1}{32} \left(\frac{x^4}{4} \right) \Big|_1^9 - \frac{1}{32} \left(\frac{x^3}{3} \right) \Big|_1^9 - 6.33^2 \end{aligned}$$

Now factor out constants, substitute the limits of integration, and simplify.

$$\begin{aligned} \frac{1}{32} \left(\frac{x^4}{4} \right) \Big|_1^9 - \frac{1}{32} \left(\frac{x^3}{3} \right) \Big|_1^9 - 6.33^2 &= \frac{1}{128} (9^4 - 1^4) - \frac{1}{96} (9^3 - 1^3) - 6.33^2 \\ &\approx 3.60 \end{aligned}$$

Therefore, the variance is $\text{Var}(X) \approx 3.60$.

Finally, find the standard deviation. The standard deviation of X is $\sigma = \sqrt{\text{Var}(X)}$.

$$\sigma = \sqrt{3.60} \approx 1.9$$